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Example 8.9. Jar of marbles: You are given a jar full of two kinds of marbles, white and black, and asked to play this game. Randomly select two marbles from the jar. If they are the same color, throw them out, but put another black marble in (you may assume that you have an endless supply of spare marbles). If they are different colors, place the white one back into the jar and throw the black one away. If you knew the original numbers of white and black marbles, what is the color of the last marble in the jar?

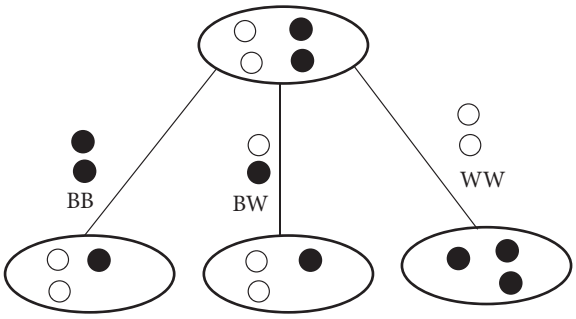

Figure 8.3: State changes in the jar marbles

The number of white and black marbles in the jar can be represented by two variables w and b. In each iterative step, b and w change depending on the colors of the two marbles taken out: Black Black, Black White or White White. It is illustrated in Figure 8.3 and annotated in the algorithm below.

```
1  while at least two marbles in jar
2     -- b , w
3     take out any two marbles
4     case both are black -- BB
5        throw away both the marbles
6        put a black marble back
7        -- b = b '-1, w = w',  b+w = b'+w' -1
8     case both are white  --WW
9        throw away both the marbles
10       put a black marble back
11       --b = b'+1, w = w'-2, --b+w = b'+w'-1
12 else                    --BW
13       throw away the black one
14       put the white one back
15       -- b = b'-1, w = w', b+w = b'+w'-1
```

For each case, how b, w and b+w change is shown in the algorithm, where b' and w' are values of the variables before taking out two marbles. Notice the way w changes. Either it does not change, or decreases by 2. This means that the parity of w, whether it is odd or even, does not change. The parity of w is invariant.

Suppose, at the start of the game, w is even. When the game ends, w is still even. Moreover, only one marble is left, w+b = 1.

1	w + b = 1	end of the loop
2	w = 0orw = 1	from 1
3	w is even	loop invariant
4	w = 0	from 2,3
5	b = 1	from 1,4

Last marble must be black. Similarly, if at the start of the game, there is an odd number of whites, the last marble must be white.

One last question: do we ever reach a state with only one marble? Yes, because the total number of marbles b+w always decreases by one at each step, it will eventually become 1.

8.4 Recursion

Recursion is an algorithm design technique, closely related to induction. It